

Construction-Conditioned Potential-Game Diagnostics for Chaperone Models

Lukas Geiger

Independent Researcher, Bernau

Code: research-line/fst-nash

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Abstract

We examine 2×2 normal-form games constructed from selected chaperone-substrate and fold-switching examples with the exact potential-game (PG) test of Monderer and Shapley (1996). Under the modeling convention used here, a shared interaction contrast is encoded as PG and an asymmetric contrast as non-PG. This convention is motivated by Xu’s (2022) symmetry framework, but it is not derived from a chaperone reaction network and therefore does not establish an $S4 \iff PG$ equivalence.

The 16-case table is consequently a *construction ledger*, not an independently validated biological atlas. XCL1/Lymphotactin provides an internal salt-shift reconstruction (23.7% Ltn40 at 150 mM NaCl) because the same Tyler et al. (2012) data set supplies the calibration and comparison. The Thermosome example shows that the same biological system can receive opposite PG labels under different game constructions. A five-axis validation ledger yields 0/8 full validation passes. Nash-equilibrium identities and sensitivity boundaries are therefore candidate outputs for future preregistered tests, not validated predictions.

Relation to Prior Work. This paper supersedes Section 3 (“Game-Theoretic Stability”) of the overview paper *Functional Stability Theory III: Biological Stability and Nash Frustration* (Geiger 2026, DOI: 10.5281/zenodo.20130573). The Nash-Frustration correlation reported there ($\rho_S = 0.44$, $p = 0.033$, $n = 24$) is not discriminatory under the earlier minimum-and-step-size construction: for a positive-definite Hessian, $\rho(I - \eta H) < 1$ follows whenever $0 < \eta < 2/\lambda_{\max}(H)$. Those results are therefore not retained as biological evidence here; the present paper uses the reformulated framework instead.

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1 Introduction

Molecular chaperones assist protein folding through diverse mechanisms ranging from passive holdase activity to ATP-driven conformational cycles. Xu [1], building on Goloubinoff et al. [2], classified chaperone mechanisms via four symmetry conditions on the rate matrix of chaperone–substrate interactions:

- S1** Detailed balance of binding/unbinding
- S2** Substrate-independent dissociation
- S3** Substrate-independent ATPase rate
- S4** Reversibility: every closed chaperone state has a reversible path to an open state without consuming chemical energy

When all four hold, the system is at equilibrium. Breaking any condition yields a non-equilibrium system. We abbreviate these as Goldilocks (GG) and non-Goldilocks (NGG) throughout; this terminology is ours, not Xu’s. We use “Goloubinoff landscape” as shorthand for the equilibrium/non-equilibrium chaperone research program founded by Goloubinoff et al. [2] and formalized into the S1–S4 conditions by Xu [1]. Xu already distinguishes Hsp70 and Hsp90 by the different subsets of symmetry conditions they break and by their different non-equilibrium mechanisms. Our narrower question is whether explicitly specified game representations can provide an auditable re-encoding of such symmetry patterns.

1.1 Contribution

We apply the *potential-game* (PG) property of Monderer and Shapley [3] to explicitly constructed 2×2 normal-form games. The four-cycle test audits a specified payoff representation; it does not independently determine Xu’s S4 condition. We use the following proposed labels as a descriptive partition:

1. **Kinetic NGG** (S3 broken, S4 treated as intact; encoded with a shared contrast, hence PG by construction): proposed for routes in which rates are substrate-dependent but no directed cycle is assumed to shift the equilibrium.
2. **Thermodynamic NGG** (S3 and S4 treated as broken; encoded with asymmetric contrasts, hence non-PG by construction): proposed for routes in which a directed thermodynamic cycle is assumed to shift the equilibrium; the present models do not compute cycle affinities.

We document 16 model constructions, including four post-hoc convention-consistency cases, reconstruct the calibrated XCL1 salt response, and illustrate construction-conditioning via the Thermosome meta-demonstration.

1.2 Relation to Prior Work

This paper replaces the Nash-Frustration analysis in FST-III [4]. That analysis modeled each residue as a player choosing backbone dihedral angles (φ_i, ψ_i) on the torus T^2 , with pairwise cosine couplings. The spectral radius $\rho(J) < 1$ of the Jacobian $J = I - \eta H$ was

interpreted as “Nash stability.” However, a random-parameter control (Section 5) showed that for a positive-definite Hessian, $\rho(I - \eta H) < 1$ follows under the step-size condition $0 < \eta < 2/\lambda_{\max}(H)$. The test is therefore not discriminatory under that minimum-and-step-size construction. Prior results, including the Spearman correlation $\rho_S = 0.44$ against B-factors, are not retained as biological evidence here.

The present framework sets aside the continuous dihedral-angle approach entirely and instead formulates chaperone–substrate interactions as discrete 2×2 normal-form games, where the PG test is a well-defined audit of the specified payoff representation.

2 Theory: Four-Cycle Test and the S4-Motivated Construction

Definition 1 (Potential Game, Monderer and Shapley 1996). *A finite n -player game is a potential game if there exists a function $\Phi : S_1 \times \cdots \times S_n \rightarrow \mathbb{R}$ such that for every player k , every strategy profile s_{-k} , and every pair of strategies s_k, s'_k :*

$$u_k(s'_k, s_{-k}) - u_k(s_k, s_{-k}) = \Phi(s'_k, s_{-k}) - \Phi(s_k, s_{-k}). \quad (1)$$

For a 2×2 game with players H (chaperone) and S (substrate), strategies $\{0, 1\}$, and payoff matrices u_H, u_S , the PG condition reduces to a single scalar test:

Proposition 1 (Four-Cycle Test for 2×2 Games). *Define the interaction index for each player k :*

$$I_k = u_k[0, 0] - u_k[0, 1] - u_k[1, 0] + u_k[1, 1]. \quad (2)$$

The game is a potential game if and only if $I_H = I_S$.

Proof. Consider the two unilateral paths from $(0, 0)$ to $(1, 1)$. Along the first path, H changes the row and S then changes the column; along the second, S changes the column and H then changes the row. A scalar exact potential is path independent if and only if

$$\begin{aligned} & [u_H(1, 0) - u_H(0, 0)] + [u_S(1, 1) - u_S(1, 0)] \\ &= [u_S(0, 1) - u_S(0, 0)] + [u_H(1, 1) - u_H(0, 1)]. \end{aligned} \quad (3)$$

Rearranging gives $I_H = I_S$, the four-cycle condition for this orientation. The converse follows by constructing Φ along either path [3]. $\square$

2.1 Construction Rule Motivated by Xu’s S4

Xu’s fourth symmetry condition (S4) [1] requires that every closed chaperone state has a reversible path to an open state without consuming chemical energy—i.e., that the chaperone–substrate reaction network is microscopically reversible.

In the present models, a case treated as S4-intact is represented with a shared interaction contrast, which makes $I_H = I_S$ by construction. A case treated as S4-broken is represented with asymmetric contrasts, which generally makes $I_H \neq I_S$. These are modeling rules, not a derivation from transition rates or a proof of biological equivalence.

Construction dependence. Potentiality is a property of the specified game. Player partition, coarse-graining, utility definition, and cardinal scale can change the PG result. A coarse-grained four-state game can also hide dissipative cycles. Thus PG does not imply biological reversibility or S4 unless a complete network-to-game mapping and its cycle affinities are independently established. The Thermosome meta-demonstration (Section 4.4) illustrates this dependence.

3 Methods

3.1 Game Construction

Each chaperone–substrate system is modeled as a 2×2 normal-form game:

- **Player H** (chaperone): strategies {Inactive/Resting, Active/Engaged}
- **Player S** (substrate): strategies {Unfolded/Free, Folded/Bound}

The payoff matrix for player $k \in \{H, S\}$ is:

$$u_k = \begin{pmatrix} u_k[0, 0] & u_k[0, 1] \\ u_k[1, 0] & u_k[1, 1] \end{pmatrix}, \quad (4)$$

where larger entries are treated computationally as preferred model scores. Some terms are energy-derived and reported in kcal/mol; other terms are transformations of kinetic ratios or assumptions. The current case models do not provide a universal thermodynamic mapping from K_D , k_{cat} , and conformational energies onto one common cardinal utility scale. Accordingly, exact-PG differences in kcal/mol are construction-level score contrasts, not direct physical measurements. A claim-bearing thermodynamic model would instead require matched forward/reverse rates or explicitly defined utilities such as $u = -G$ at fixed temperature and standard state.

The coupling energy Δ_k represents the change in player k ’s payoff when the other player switches strategy:

$$\Delta_k = u_k[\text{coupled}] - u_k[\text{uncoupled}]. \quad (5)$$

3.2 Four-Cycle Test (PG Test)

For each calibrated system, we compute I_H and I_S via Eq. (2) and evaluate:

$$\text{PG violation} = |I_H - I_S|. \quad (6)$$

If the violation is below a numerical threshold ($\varepsilon = 10^{-10}$), the specified game is classified as an exact PG. This numerical identity is not an empirical S4 decision and does not propagate measurement uncertainty.

3.3 Nash Equilibrium Search

Pure Nash equilibria are identified by checking all four strategy profiles for unilateral deviation incentives. A profile (s_H^*, s_S^*) is a pure NE if:

$$u_H(s_H^*, s_S^*) \geq u_H(s_H', s_S^*) \quad \forall s_H', \quad (7)$$

$$u_S(s_H^*, s_S^*) \geq u_S(s_H^*, s_S') \quad \forall s_S'. \quad (8)$$

3.4 Regime Classification

Based on PG test and Xu’s symmetry conditions [1]:

1. **GG** (Equilibrium): S1–S4 treated as intact; the chosen shared-contrast representation is PG; no ATP or no modeled substrate stimulation

2. **Kinetic NGG:** S3 broken, S4 treated as intact; the chosen shared-contrast representation is PG
3. **Thermodynamic NGG:** S3 and S4 treated as broken; the chosen asymmetric-contrast representation is non-PG
4. **Special cases:** non-standard 2×2 structure (3-player, oscillatory, regulated)

3.5 Symmetry Mapping

For each system, we evaluate S1–S4 based on published experimental data:

- S1: Does the system obey detailed balance for binding/unbinding?
- S2: Is dissociation substrate-independent?
- S3: Is the ATPase rate substrate-independent?
- S4: Does the cited biological route meet Xu’s reversibility condition? This label is recorded separately from the game result; the present construction encodes an S4-intact label with a shared contrast and therefore with $I_H = I_S$.

4 Results

4.1 Construction-Conditioned Regime Proposal

The 16 game constructions can be organized into three proposed labels (Table 1). This is a descriptive grouping of the chosen representations, not an independently learned biological partition:

1. **Equilibrium (GG):** 6 systems. PG = True, no S3-breaking. Includes XCL1 (metamorphic fold-switching), Prefoldin (holdase), and 4 post-hoc convention-consistency systems (DnaJ, Hsp26, Hsp33, Hsc70).
2. **Kinetic NGG:** 2 systems (Hsp70/DnaJ, SecA). Xu identifies Hsp70-mediated folding as breaking S3. His necessary condition for breaking S4 requires at least two ATP molecules to be hydrolyzed sequentially per cycle, which is absent from the Hsp70 route considered here. The biological S4 label is therefore consistent with Xu. Separately, PG=True in our matrix follows from the assumed shared interaction contrast; it is not an independent test of that label.
3. **Thermodynamic NGG:** 4 systems (Hsp90/Cdc37, GroEL/GroES, ClpB/Hsp104, p97/VCP). PG = False with constructed score differences of 0.5–2.5. These systems are represented as breaking both S3 and S4. The motivating hypothesis is that co-chaperones, multi-ring architectures, or domain asymmetries create directed thermodynamic cycles that shift the equilibrium; the present 2×2 scores encode this assumption and do not compute or test cycle affinities. For GroEL, substrate release requires sequential ATP hydrolysis in the trans-ring, violating Xu’s S4 [1].

Four additional systems (Trigger Factor, KaiB/KaiC, Hsp70→Hsp90 handoff, MAD2) fall outside the standard three-regime classification due to non-standard game structures (ribosome-coupled, oscillatory, 3-player, or regulated metamorphic).

4.2 Construction Ledger

Table 1: Construction Ledger: 16 specified game representations audited with the exact-PG test. The final column records input status and the construction that determines PG or non-PG. No row is an independent validation of a biological label (see Section 4.4). NE = number of pure Nash equilibria in the specified game.

#	System	ATP?	PG	NE	Regime	Evidence / PG basis
<i>Regime 1 — Equilibrium (GG)</i>						
1	XCL1 (metamorphic)	No	✓	2	GG	Symmetric game by construction; calibrated.
2	Prefoldin (holdase)	No	✓*	2*	GG*	Estimated parameters ($f=0.5$); not independently calibrated.
3	DnaJ alone	No	✓	1	GG	Shared- Δ convention; post-hoc consistency case.
4	Hsp26 sHsp	No	✓	1	GG	Shared- Δ convention; post-hoc consistency case.
5	Hsp33 redox	No	✓	1	GG	Shared- Δ convention; post-hoc consistency case.
6	Hsc70 alone	Basal	✓	1	GG	Shared- Δ convention; post-hoc consistency case.
<i>Regime 2 — Kinetic NGG</i>						
7	Hsp70/DnaJ	Yes	✓	1	kin. NGG	Shared- Δ from experimental K_D values; calibrated.
8	SecA	Yes	✓	1	kin. NGG	Shared- Δ from experimental K_D values; calibrated.
<i>Regime 3 — Thermodynamic NGG</i>						
9	Hsp90/Cdc37	Yes	×	1	therm. NGG	Asymmetric Δ (calibrated/estimated); calibrated.
10	GroEL/GroES	Yes	×	1	therm. NGG	Experimentally motivated ring asymmetry; calibrated.
11	ClpB/Hsp104	Yes	×	1	therm. NGG	Threading asymmetry; calibrated.
12	p97/VCP	Yes	×	1	therm. NGG	Estimated D1/D2 asymmetry; calibrated.
<i>Special cases (non-standard regime)</i>						
13	Trigger Factor	No	×	1	non-GG	Ribosome asymmetry using experimental K_D ; calibrated.
14	KaiB/KaiC	Indirect	×	1	non-GG	Oscillatory system with experimental kinetics; calibrated.
15	Hsp70→Hsp90 (3-player)	Yes	×	1	non-PG	Three-player model, not a 2×2 game; calibrated.
16	MAD2 (regulated metamorphic)	No [‡]	×	1	non-PG	TRIP13-regulated asymmetry; calibrated.

Honesty note. All PG=True systems have PG as an algebraic consequence of the modeling convention: XCL1 from a symmetric game (identical players), Hsp70/DnaJ and SecA from the assumption $\Delta_H = \Delta_S$ (experimental K_D values, but symmetric coupling is *assumed*), and post-hoc cases from the shared- Δ convention. All non-PG systems have non-PG from $\Delta_H \neq \Delta_S$ or from asymmetric game structure. PG/non-PG is thus a *convention decision during modeling*, not an emergent empirical result. * Prefoldin is based on estimated parameters (K_D , ΔG_{conf} unpublished). [‡] MAD2 has no direct ATP; TRIP13 AAA+

ATPase resets MAD2.

4.3 XCL1 Salt-Shift: Internal Calibrated Reconstruction

XCL1/Lymphotactin is a metamorphic protein that interconverts between a chemokine fold (Ltn10) and a β -sheet fold (Ltn40) with $K_{\text{eq}} = 1.2$ at 25 °C, 0 mM NaCl as the project’s calibration anchor. The internal records refer to both Tuinstra et al. [5] and Tyler et al. [6]; the precise primary-text attribution of this value remains unverified. We model XCL1 as a symmetric 2×2 coordination game between two identical monomers, with strategies {Chemokine, Beta-sheet}.

The game yields 2 pure NE: (Chem, Chem) = Ltn10 and (Beta, Beta) = Ltn40, representing the metamorphic bistability as a coordination equilibrium.

4.3.1 Salt-Dependent Population Shift

Tyler et al. [6] showed that NaCl shifts the population from $\sim 55\%$ Ltn40 (0 mM) to $\sim 24\%$ Ltn40 (150 mM) via chloride binding at Arg23/Arg43 ($K_d \approx 16$ mM from CSP titration). We model this as a chloride-binding saturation:

$$f(\text{Cl}^-) = \frac{[\text{Cl}^-]}{[\text{Cl}^-] + K_d}, \quad \Delta\Delta G([\text{NaCl}]) = \Delta\Delta G_{150} \frac{f([\text{NaCl}])}{f(150 \text{ mM})}, \quad (9)$$

with $\Delta\Delta G_{150} = 0.8$ kcal/mol (Tyler 2012 ZZ-exchange: $\Delta\Delta G(k_f) = 0.5$ and $\Delta\Delta G(k_r) = 0.3$ kcal/mol, summing to 0.8). Thus 0.8 kcal/mol is the encoded 150-mM anchor, not the asymptotic shift; the implied asymptote is approximately 0.885 kcal/mol. This normalization matches the implementation and table. Its precise source semantics and uncertainty still require a primary-full-text audit.

Table 2: XCL1 Salt-Shift Reconstruction. Model calibrated at $K_{\text{eq}}(0 \text{ mM}) = 1.2$, $\Delta\Delta G_{150} = 0.8$ kcal/mol, and $K_d = 16$ mM.

[NaCl] (mM)	$f(\text{Cl}^-)$	K_{eq}	% Ltn40	Experimental
0	0.00	1.200	54.5	54.5% (project anchor)
10	0.39	0.676	40.3	—
50	0.76	0.387	27.9	—
75	0.82	0.351	26.0	Tyler: “similar to 150 mM”
150	0.90	0.311	23.7	$\sim 24\%$ Tyler ZZ-exch.
300	0.95	0.291	22.5	—
365	0.96	0.287	22.3	CSP saturation (Tyler)

4.3.2 Independence Status

The 150 mM reconstruction (23.7%) uses Tyler’s $\Delta\Delta G$ as input and recalculates the same evidence family. It is not an independent prediction. The remaining internal candidate outputs are:

1. **75 mM grid point:** The model gives 26.0%. Tyler’s qualitative statement that 70 mM and 150 mM had a similar effect comes from the calibration evidence family and is not an external test.

2. **Saturation curve shape:** The nonlinear form is determined by chloride-binding saturation ($K_d = 16$ mM from CSP), not derivable from Tyler’s population data alone.
3. **K25A mutant:** The model output is $K_{eq} = 0.40$ (28.6% Ltn40). It is not treated here as an independently preregistered mutant prediction.

4.4 Construction-Conditioning: Thermosome Meta-Demonstration

To demonstrate that PG/non-PG depends on the game formulation and is not fixed by the biological system alone, we analyze the Thermosome of *Thermoplasma acidophilum*—a Group II chaperonin with $(\alpha\beta)_4(\alpha\beta)_4$ hexadecameric architecture and negative inter-ring cooperativity [7, 8]—using two deliberately contrasting game encodings:

Table 3: Thermosome: Same Biological System, Two Game Models, Opposite PG Results.

Model	Players	PG	NE	I_1	I_2	Regime
A: Ring vs. Ring	$\alpha_4\beta_4$ -Ring 1 vs. Ring 2	✓	2	−12.92	−12.92	GG
B: Chaperone vs. Substrate	Hexadecamer vs. Substrate	×	1	−4.0	−1.5	therm. NGG

Model A treats the two rings as identical players in a symmetric anti-coordination game {Tight, Relaxed}. PG follows algebraically because $u_1 = u_2^\top$ (identical players $\Rightarrow I_1 = I_2$ by construction). Its two static NE are (Tight, Relaxed) and (Relaxed, Tight). They represent complementary ring states but do not by themselves establish temporal alternation; that would require a transition or Markov model.

Model B treats the hexadecamer and substrate as asymmetric players. It converts an ATP-binding input and a one-way substrate-folding acceleration into score terms $\Delta_H = 4.0$ and $\Delta_S = 1.5$. Because those observables do not by themselves define a shared thermodynamic scale, their 2.5 difference is a constructed PG-score contrast rather than a measured free-energy violation.

The pair illustrates model dependence, not two equally validated biological mechanisms. The choice is a modeling decision and motivates construction-conditioning transparency throughout this paper.

4.4.1 Robustness

A scan over $\Delta_H \in [0.5, 8.0] \times \Delta_S \in [0.5, 8.0]$ confirms that Model B yields PG=True exclusively on the diagonal $\Delta_H = \Delta_S$, as follows algebraically from the payoff structure. This is sensitivity to the specified score scan; the scan does not establish which parameter combinations are physically plausible.

5 Known Limitations and Tautology Transparency

5.1 What Is Tautological

1. **PG/non-PG follows from payoff construction:** $\Delta_H = \Delta_S \Rightarrow$ PG (algebraically); $\Delta_H \neq \Delta_S \Rightarrow$ non-PG (algebraically). The PG test tests the payoff symmetry, not a

system property.

2. **NE multiplicity:** In a generic (tie-free) 2×2 exact potential game, two pure NE occur if and only if neither player has a strictly dominant strategy; this is a consequence of the game structure, not a biological finding.
3. **Prior Nash-Frustration results (FST-III v3):** The spectral radius $\rho(I - \eta H) < 1$ at a non-degenerate minimum with positive-definite Hessian follows for $0 < \eta < 2/\lambda_{\max}(H)$. Under that construction it is not a discriminator of biological stability. Random parameters satisfy the same criterion ($\rho(J) < 1$ in 50/50 random and 10/10 fitted runs).

Scale limitation. Exact potentiality is a cardinal property. Separate positive rescalings of player utilities can preserve best responses and Nash equilibria while changing $I_H = I_S$. A common physical scale must therefore be derived before exact PG is interpreted thermodynamically; weighted and ordinal potentials answer weaker, different questions.

5.2 Candidate Outputs Not Yet Independently Validated

1. **NE identity and sensitivity:** The specified matrices produce equilibrium profiles and score-boundaries that can be frozen for future tests.
2. **Salt response:** The calibrated XCL1 reconstruction produces a concentration-response curve, but no untouched concentration measurement has yet been tested after model freeze.
3. **Regime labels:** The proposed grouping is an auditable re-encoding of selected symmetry patterns, not an empirically discovered ontology.

5.3 Pre-Review Validation-Evidence Gate

The machine-readable validation ledger generated by `scripts/validation_evidence_ledger.py` separates eight headline case routes across five axes: construction conditioning, experimental dynamics, an AlphaFold baseline, an energetic-frustration baseline, and an independent chaperone-cycle anchor. All eight rows are construction-conditioned; none contains a computed AlphaFold baseline, an energetic-frustration baseline, or a quantitative independent cycle anchor. Consequently, the full-stack validation count is 0/8.

This gate does not erase the narrower XCL1 population consistency check, but it preserves its partial-circularity caveat. GroEL remains only a qualitative partial anchor: two of three kinetic categories are consistent, while all modeled cycle times miss the stated 15–20 s experimental range (primary source pending verification). The current evidence therefore supports diagnostic classification only, not a validated mechanistic model or a claim upgrade.

6 Discussion

6.1 Honest Balance

We evaluate our framework against 13 open questions in the chaperone field (Table 4). The framework addresses $\sim 38\%$ of questions (5/13), of which only one (Q9, XCL1 salt-shift) has a quantitative, partially circular consistency check.

Table 4: Honest Balance: Framework Coverage of 13 Open Questions.

Category	Questions	Fraction
QUANTITATIVE CHECK (partly circular)	Q9 (XCL1 salt-shift)	7.7%
MODEL OUTPUT (construction-conditioned)	Q1, Q6, Q10, Q13	30.8%
RE-DESCRIBE	Q4, Q8	15.4%
NONE	Q2, Q3, Q5, Q7, Q11, Q12	46.2%

~46% of questions lie entirely outside the framework’s scope. This balance is the honest foundation of this paper.

6.2 Comparison with Existing Approaches

The PG test provides a *diagnostic* classification layer that complements rather than replaces existing approaches:

- **Statistical mechanics (Ising, MWC):** If the exact potential of a PG game is identified with a negative free energy on a common physical scale, pure-NE search reduces to locating free-energy minima under unilateral moves and adds nothing beyond statistical mechanics. Within a specified cardinal representation, PG is necessary but not sufficient for this reduction; the present scores do not establish such a scale (Section 3.1).
- **AlphaFold / structural prediction:** Our framework operates on a different level—it labels a specified game representation of conformational regimes, not the structure itself. For metamorphic proteins (XCL1, MAD2), the NE analysis supplies construction-conditioned candidate descriptions of bistability.
- **Xu/Goloubinoff framework:** Xu already distinguishes mechanisms by their broken symmetry subsets. Our contribution is a proposed game-theoretic re-encoding and audit, not the discovery of that biological distinction.

6.3 What This Paper Does Not Claim

- Nash equilibrium “explains” chaperone function (it classifies).
- PG/non-PG is an “explanation” for a phenomenon (it is construction-conditioned).
- “Three regimes instead of two” as an extension of Goloubinoff (it is substructure).
- AF2 failure is “solved” by Nash analysis (it is reformulated).
- ATP costs are “predicted” by Nash analysis (Goloubinoff already explains them).
- The 150 mM population is an independent prediction (it is a calibrated internal reconstruction).

7 Conclusion

The four-cycle test is mathematically valid for a specified game, but its biological interpretation is conditional on player partition, coarse-graining, utility definition, and cardinal scale. Current evidence does not establish an $S4 \iff PG$ equivalence, a validated 16-system atlas, or an independent XCL1 prediction. The present contribution is a construction ledger with explicit failure modes and a 0/8 validation status.

The next claim-bearing step is a preregistered network-to-game construction from matched forward and reverse rates, with uncertainty propagation, a hidden-cycle audit, and a blinded external endpoint.

All scripts (36 Python files) and result files are available at [GitHub: research-line/fst-nash](https://github.com/research-line/fst-nash).

Data and Code Availability

All computational scripts are located in `scripts/` (36 files) with results in `results/` (JSON format). The template functions for the four-cycle test, Nash-equilibrium search, and symmetry mapping are in `scripts/fold_switch_template.py`. Hold-out scripts are in `scripts/extension_B/`. The pre-review evidence gate is generated by `scripts/validation_evidence_ledger.py`.

AI Disclosure

AI systems were used extensively in all phases of this work: conception and research discussion, literature and source work, mathematical analysis and proof development, creation and checking of the computational scripts, execution and evaluation of the computational runs, text, translation, and typesetting.

Quality assurance: use of several independent models from different providers with counter-checks and adversarial AI reviews; traceable, versioned computational scripts with stored result files and checksums; continuously maintained proof and register documents; strict claim hygiene (assertions only with documented evidence, addenda instead of silent corrections); changelogs across all versions.

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